

Partial Tax Recovery on Returns (α -Reclaim under Seller and Buyer Remittance)

Draft notes
Companion to `welfare_with_returns`

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1 Setup

The companion note studies the extreme in which the remitter recovers none of the unit tax τ from the government when a unit is returned. Here the remitter recovers a fraction $\alpha \in [0, 1]$ of τ on each returned unit, whether the seller or the buyer remits. Full reclaim is the boundary $\alpha = 1$; zero reclaim is $\alpha = 0$. Private refunds between buyer and seller are unchanged from the companion note. Under seller remittance the seller refunds the tax-inclusive checkout price p . Under buyer remittance the seller refunds only the pre-tax list price $p - \tau$, and the buyer reclaims $\alpha\tau$ from the treasury on a return.

Demand $D(p)$, return rate $r(\cdot)$, handling h , and market supply $S(\mu)$ are as in the companion note. Equilibrium clears at $D(p) = S(\mu)$, with μ the remitter- and α -dependent effective margin derived below. Write $Q \equiv D(p)$ and evaluate local derivatives at $\tau = 0$, where checkout and list prices coincide.

2 Effective margins

Under seller remittance the seller remits τ on every gross sale and recovers $\alpha\tau$ from the government on each return. The private refund remains p , so the return rate is still $r(p)$. Expected net receipt per gross unit is therefore

$$\mu^s(p, \tau; \alpha) = p - \tau - r(p)(p + h - \alpha\tau). \quad (1)$$

The term $-\alpha\tau$ inside the return cost is the treasury reclaim. At $\alpha = 0$ this is the companion-note margin $p - \tau - r(p)(p + h)$.

Under buyer remittance the seller never remits τ . The buyer remits on purchase and recovers $\alpha\tau$ on a return, while the seller refunds only the pre-tax price $p - \tau$. The buyer's total recovery on a return is therefore

$$(p - \tau) + \alpha\tau = p - (1 - \alpha)\tau,$$

so the return rate is evaluated at $p - (1 - \alpha)\tau$. The seller's effective margin is

$$\mu^b(p, \tau; \alpha) = (p - \tau) - r(p - (1 - \alpha)\tau)(p - \tau + h). \quad (2)$$

At $\alpha = 0$ the argument of r collapses to $p - \tau$ and (2) recovers the companion note. Clearing conditions are $D(p) = S(\mu^s(p, \tau; \alpha))$ and $D(p) = S(\mu^b(p, \tau; \alpha))$.

3 Tax neutrality

At a common checkout price p and tax τ , the two regimes define the same equilibrium if and only if $\mu^s = \mu^b$. Canceling $p - \tau$ gives

$$r(p)(p + h - \alpha\tau) = r(p - (1 - \alpha)\tau)(p - \tau + h). \quad (3)$$

Result 1 (Partial recovery and tax neutrality). *Seller and buyer remittance are equivalent if and only if (3) holds. When $\alpha = 1$, both sides equal $r(p)(p - \tau + h)$, so neutrality holds for every r . When $\alpha \in [0, 1)$, neutrality fails for generic r , as in the zero-reclaim companion note.*

Full reclaim undoes the remittance wedge on returns. Any incomplete reclaim leaves a wedge between the seller's return cost and the buyer's keep-or-return price, so which party remits continues to matter.

4 Pass-through at $\tau = 0$

Write $\mu_p \equiv 1 - r(p) - r'(p)(p + h)$ for the no-tax response of the effective margin to list price, and let ρ_c denote production-cost pass-through as in the companion note. Differentiating (1)–(2) at fixed p and evaluating at $\tau = 0$,

$$\mu_\tau^s = -(1 - \alpha r(p)), \quad \mu_\tau^b = -\mu_p - \alpha r'(p)(p + h). \quad (4)$$

At $\tau = 0$ the seller-side list-price derivative remains μ_p , because the reclaim term $\alpha\tau$ vanishes from $\partial\mu^s/\partial p$ when $\tau = 0$. Pass-through into checkout price is therefore

$$\rho_\tau^s = (1 - \alpha r(p)) \rho_c, \quad \rho_\tau^b = (\mu_p + \alpha r'(p)(p + h)) \rho_c. \quad (5)$$

When $\alpha = 0$, (5) collapses to $\rho_\tau^s = \rho_c$ and $\rho_\tau^b = \mu_p \rho_c$. A larger reclaim rate shrinks the seller-tax wedge by the factor $1 - \alpha r(p)$: only the unclaimed share of expected tax $(1 - \alpha r)\tau$ acts like a resource cost on the supply side. Under buyer remittance, reclaim shifts the return-rate argument and adds the term $\alpha r'(p)(p + h)$ to the companion-note factor μ_p .

5 Revenue and first-order surplus

On each gross sale the treasury collects τ . On each return it refunds $\alpha\tau$ to the remitter. Net revenue is therefore

$$G = \tau Q(1 - \alpha r^*), \quad (6)$$

where $r^* = r(p)$ under seller remittance and $r^* = r(p - (1 - \alpha)\tau)$ under buyer remittance. At $\tau = 0$ both give $dG/d\tau = Q(1 - \alpha r(p))$.

Private surplus $W = CS + PS$ is differentiated as in the companion note, using $dCS/d\tau = -Q(1 - r)\rho_\tau$ and $dPS/d\tau = Q(\mu_p \rho_\tau + \mu_\tau)$ at $\tau = 0$. Substituting (4)–(5) and writing $\Lambda \equiv 1 + r'(p)(p + h)\rho_c$ yields

$$\frac{dW}{d\tau^s} = -Q(1 - \alpha r(p)) \Lambda, \quad \frac{dW}{d\tau^b} = -Q(\mu_p + \alpha r'(p)(p + h)) \Lambda. \quad (7)$$

Social surplus $V = W + G$ then satisfies, at $\tau = 0$,

$$\frac{dV}{d\tau^s} = -Q(1 - \alpha r(p)) (\Lambda - 1), \quad \frac{dV}{d\tau^b} = Q[(1 - \alpha r(p)) - (\mu_p + \alpha r'(p)(p + h)) \Lambda]. \quad (8)$$

When $\alpha = 0$, (7)–(8) reproduce the companion note. When $\alpha \rightarrow 1$, seller remittance has $dW/d\tau^s \rightarrow -Q(1-r)\Lambda$ and $dV/d\tau^s \rightarrow -Q(1-r)(\Lambda-1)$, so the unclaimed tax on returned units vanishes from the first-order social wedge and only the kept-unit share $1-r$ remains in the private loss. Incomplete reclaim $\alpha < 1$ leaves a strictly larger seller-side resource cost $1-\alpha r > 1-r$ whenever $r \in (0,1)$.

Result 2 (Partial recovery nests zero and full reclaim). *For $\alpha \in [0,1]$, equations (1)–(8) nest the zero-reclaim companion note at $\alpha = 0$. At $\alpha = 1$, tax neutrality is restored and the seller-tax pass-through factor collapses to the keep rate $1-r(p)$.*